FLOQUET SPARSE OBSTRUCTION: EXACT EXCLUSION THROUGH SUPPORT SIX IN THE PERIOD-THREE HAMMING FIBRE

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Abstract. Let Δ be the adjacency-form discrete Laplacian on $\mathbf{Z}^d$, and let $V\colon \mathbf{Z}^d \to \mathbf{C}$ be $(3\mathbf{Z})^d$-periodic. We study complex potentials for which $\Delta + V$ and Δ are Floquet isospectral. At zero quasimomentum the Floquet matrix is the adjacency matrix of the Hamming graph $H(d,3)$. Trace identities at this single fibre show, by a completely human-checkable argument, that every nonzero isospectral diagonal shift of this fibre has at least five nonzero entries for every $d \geq 1$. For $d \geq 3$, equality of the first five traces on a support of size at most four has a unique nonzero form: an induced claw with centre value -1 and leaf values $1, i, -i$. Its sixth trace has the nonzero defect $-12(2d-5)$, so the sixth moment is sharp for this sparse trace test.

For the smallest unresolved rectangular period, $(3\mathbf{Z})^3$, we strengthen the lower bound to seven by an exact computer-assisted proof. The resolvent of $H(3,3)$ depends only on Hamming distance. The matrix determinant lemma therefore reduces supports of sizes five and six to, respectively, 104 and 320 distance-matrix types. After imposing exact support by Rabinowitsch extension, every resulting extension ideal has reduced Gröbner basis $\{1\}$ over $\mathbf{Q}$. The computation uses exact rational polynomial arithmetic; an independent orbit and Burnside audit verifies the completeness of the type lists, and a deterministic transcript digest records every type. Thus every nonzero shift of support at most six is detected already at the periodic fibre. This closes the support-at-most-six stratum of the open $(3,3,3)$ problem, but does not decide whether a nonzero potential of unrestricted support exists.

1. Introduction

For $n \in \mathbf{Z}^d$ put

$$(\Delta f)(n) = \sum_{\|n-m\|_1=1} f(m).$$

Let V be $(3\mathbf{Z})^d$-periodic and identify a fundamental cell with $\mathcal{F}_d = \{0,1,2\}^d$. Put $\mathbf{T} = \{z \in \mathbf{C} : |z| = 1\}$. For $z = (z_1, \dots, z_d) \in \mathbf{T}^d$, the Floquet matrix

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$L_V(z)$ is defined by

$$(L_V(z)f)(n) = V(n)f(n) + \sum_{j=1}^{d} (\eta_j^+(n)f(n+e_j) + \eta_j^-(n)f(n-e_j)),$$

where the arguments of f are reduced modulo 3 and

$$\eta_j^+(n) = \begin{cases} z_j, & n_j = 2, \\ 1, & n_j \neq 2, \end{cases} \qquad \eta_j^-(n) = \begin{cases} z_j^{-1}, & n_j = 0, \\ 1, & n_j \neq 0. \end{cases}$$

We say that V is *Floquet isospectral to zero* when

$$\det(xI - L_V(z)) = \det(xI - L_0(z)) \qquad (z \in \mathbf{T}^d). \tag{1}$$

The trace and spectral-invariant approach to discrete periodic inverse problems goes back at least to Kappeler [5]; see also Liu [6], Saburova's complete invariant system for periodic graphs [7], and the algebraic framework surveyed by Shipman and Sottile [8]. Real-valued potentials isospectral to zero are rigid, whereas nonzero complex examples occur for several larger periods. Faust, Liu, Matos, Plute, Robinson, Tao, Tran and Zhuang gave explicit complex examples, including a support-four construction with values $1+i, 1-i, -1+i, -1-i$ for even periods at least four [4].

Cobb, Faust and Kretschmer proved existence when at least one rectangular period exceeds 3, and rigidity when all periods are at most 3 and at most two of them equal 3 [1]. Thus $(3,3,3)$ is the smallest rectangular period not covered by their classification; they also record that generating its full 27-variable spectral-invariant ideal was beyond the direct methods they attempted. The general period-classification problem and the all-period-three barrier are stated explicitly in [3, Problem 1].

This paper does not solve the unrestricted $(3,3,3)$ case. It isolates a sparse stratum and replaces the full ideal by exact low-rank determinant identities at one necessary Floquet fibre. The result is an exact exclusion through support six, together with a finite decision procedure for every fixed Hamming-fibre support size. Here "exact" refers to exhaustive type enumeration and rational polynomial arithmetic; it does not assert that a support-seven solution exists.

Theorem 1.1 (Human Hamming-fibre obstruction)**.** *Let $d \geq 1$, let $A_d = A(H(d,3))$, and let D be a complex diagonal matrix. If $A_d + D$ and A_d have the same characteristic polynomial, then*

$$D \neq 0 \quad \Longrightarrow \quad |\operatorname{supp} D| \geq 5.$$

Theorem 1.2 (Exact $H(3,3)$ obstruction)**.** *Let $A = A(H(3,3))$ and let D be a complex diagonal matrix. If $A + D$ and A have the same characteristic polynomial, then*

$$D \neq 0 \quad \Longrightarrow \quad \boxed{|\operatorname{supp} D| \geq 7}.$$

The exclusion of supports five and six is computer-assisted and uses exact arithmetic over $\mathbf{Q}$.

Corollary 1.3 (Floquet consequence)**.** *If a $(3\mathbf{Z})^3$-periodic complex potential V is Floquet isospectral to zero, then $V = 0$ or its support in a fundamental cell has size at least seven.*

Corollary 1.4 (Non-vacuity at the periodic fibre)**.** *There exists a nonzero complex diagonal matrix D such that $A(H(3,3)) + D$ and $A(H(3,3))$ have the same characteristic polynomial, and every such D has support at least seven.*

Proof. The graph $H(3,3)$ is neither edgeless nor complete. The fixed-matrix existence theorem of Cobb–Faust–Kretschmer [1, Corollary B] therefore supplies a nonzero isospectral diagonal shift. Its support bound is Theorem 1.2. This fixed-fibre existence statement does not imply simultaneous isospectrality at all Floquet phases. □

The proof of Theorem 1.1 uses only traces through order six and a nine-row graph table. Theorem 1.2 uses the determinant lemma, the Hamming association scheme and Rabinowitsch-extended coefficient ideals. The calculation is finite because the zero-quasimomentum resolvent is a function of pairwise Hamming distance. No floating-point arithmetic, numerical root test or tolerance enters the calculation.

We are not aware of a previously stated support lower bound of this form. That literature assessment is not a priority certificate; the claim should be revisited after expert feedback.

2. The zero Floquet fibre

Write

$$\mathcal{F}_d = (\mathbf{Z}/3\mathbf{Z})^d.$$

At $z = (1, \dots, 1)$ the free Floquet matrix is

$$A_d = A(H(d,3)), \qquad H(d,3) = K_3 \,\square\, \cdots \,\square\, K_3.$$

For a diagonal matrix D indexed by $\mathcal{F}_d$, set

$$\operatorname{supp} D = \{a \in \mathcal{F}_d : D_{aa} \neq 0\}.$$

Let D be any complex diagonal matrix such that $A_d + D$ and A_d have the same characteristic polynomial. Newton identities then give

$$\operatorname{tr}(A_d + D)^k = \operatorname{tr} A_d^k \qquad (k \geq 1). \tag{2}$$

For a Floquet-isospectral potential one takes $D = \operatorname{diag}(V|_{\mathcal{F}_d})$; equation (1) implies the required finite-matrix hypothesis. Only this necessary zero-quasimomentum condition will be used. These power traces are the single-fibre specialization of the spectral invariants studied more generally in [7].

The following elementary local identities are immediate from the product graph:

$$(A_d)_{xx} = 0, \qquad (A_d^2)_{xx} = 2d, \qquad (A_d^3)_{xx} = 2d, \tag{3}$$

and

$$(A_d^2)_{xy} = 1 \qquad (x \sim y). \tag{4}$$

Every edge lies in a unique triangle: its endpoints differ in one coordinate, and the third symbol in that coordinate gives the unique common neighbour adjacent to both within the same coordinate fibre.

Lemma 2.1 (The first three moments)**.** *Let $v_1, \dots, v_r$ be the nonzero entries of D, and put $p_j = \sum_{a=1}^{r} v_a^j$. If (2) holds for $k = 1, 2, 3$, then*

$$p_1 = p_2 = p_3 = 0.$$

Proof. The first trace gives $p_1 = 0$. Since A_d has zero diagonal,

$$\operatorname{tr}(A_d + D)^2 - \operatorname{tr} A_d^2 = p_2.$$

For the third power, cyclicity of trace and (3) give

$$\operatorname{tr}(A_d + D)^3 - \operatorname{tr} A_d^3 = p_3 + 3\operatorname{tr}(A_d^2 D) = p_3 + 6dp_1 = p_3.$$

□

Corollary 2.2 (Value-level obstruction)**.** *Every nonzero isospectral diagonal shift of A_d assumes at least four distinct nonzero values on its support.*

Proof. Suppose the distinct nonzero values are $a_1, \dots, a_s$, with positive multiplicities $m_1, \dots, m_s$, where $s \le 3$. Lemma 2.1 gives

$$\sum_{j=1}^{s} m_j a_j^k = 0 \qquad (k = 1, 2, 3).$$

The first s equations have coefficient matrix $(a_j^k)_{1 \le k, j \le s}$, whose determinant is a nonzero Vandermonde factor because the a_j are nonzero and distinct. Hence every $m_j = 0$, a contradiction. □

3. A human proof through support four

Lemma 3.1 (Newton reduction)**.** *No nonzero isospectral diagonal shift of A_d has support at most three. If its support has size four, then for some $\alpha \in \mathbf{C}^\times$ its four values are*

$$\alpha\{1, -1, i, -i\}. \tag{5}$$

Proof. Newton's identities and Lemma 2.1 give $e_1 = e_2 = e_3 = 0$ for the elementary symmetric functions of the nonzero values. If $r \le 3$, the product e_r is therefore zero, contrary to the definition of the support. If $r = 4$, the monic polynomial with roots $v_1, \dots, v_4$ has the form $X^4 + e_4$, which yields (5) after choosing a fourth root. □

Let S be a four-point support and write $v_x = \alpha\omega_x$, where the labels on S are the four fourth roots of unity. From the graph induced on S define

$$E = \sum_{\{x,y\} \in E(S)} \omega_x \omega_y, \qquad T = \sum_{\{x,y\} \in E(S)} \omega_x \omega_y (\omega_x + \omega_y). \tag{6}$$

Lemma 3.2 (Trace four and trace five)**.** *Assume that D has four-point support and write its values as $v_x = \alpha\omega_x$ as above. Under the first-three-moment relations,*

$$\operatorname{tr}(A_d + D)^4 - \operatorname{tr} A_d^4 = 4\alpha^2(E + \alpha^2), \tag{7}$$

$$\operatorname{tr}(A_d + D)^5 - \operatorname{tr} A_d^5 = 5\alpha^2(2E + \alpha T). \tag{8}$$

Consequently a four-point candidate must satisfy

$$E \neq 0, \qquad T^2 + 4E = 0, \qquad \alpha = -\frac{2E}{T}. \tag{9}$$

Proof. The graph is vertex-transitive, so $(A_d^j)_{xx}$ is independent of x; hence every cyclic word containing exactly one D is a multiple of p_1. At order four, cyclic word classes give

$$\operatorname{tr}(A_d + D)^4 - \operatorname{tr} A_d^4 = \operatorname{tr} D^4 + 2\operatorname{tr}(A_d D A_d D),$$

after the terms containing p_1, p_2, p_3 are removed. Since $\operatorname{tr} D^4 = 4\alpha^4$ and $\operatorname{tr}(A_d D A_d D) = 2\alpha^2 E$, this is (7). At order five, the only surviving cyclic classes are $A_d^2 D A_d D$ and $A_d D A_d D^2$. Using $(A_d^2)_{xy} = 1$ on an edge, their total contribution is

$$10\alpha^2 E + 5\alpha^3 T,$$

while $\operatorname{tr} D^5 = \alpha^5 \sum \omega_x = 0$. This is (8). If both differences vanish, then $\alpha^2 = -E$, so $E \neq 0$, and $\alpha T = -2E$. Eliminating α gives (9). □

There are eleven unlabelled simple graphs on four vertices. The unique triangle property excludes K_4 and the diamond. Fixing one root label to 1 leaves six permutations of the other labels. This normalization is lossless: for any fourth root ζ, replacing (α, ω_x) by $(\alpha/\zeta, \zeta\omega_x)$ preserves every value v_x and the trace conditions. Direct substitution in (9) gives the complete table below.

TABLE 1. The complete trace-four/five test after normalizing one root label to 1. Graphs K_4 and the diamond cannot occur in $H(d, 3)$.

Induced graph	labelings with $E \neq 0$	survive traces $4, 5$
$4K_1$	0	0
$K_2 + 2K_1$	6	0
$P_3 + K_1$	4	0
$K_3 + K_1$	6	0
$K_{1,3}$	6	6
$2K_2$	4	0
P_4	6	0
paw	4	0
C_4	4	0

For the claw, let c be the root label at the centre. The leaf labels are the other three roots, whence

$$E = -c^2, \qquad T = -2c^3.$$

The trace equations force $\alpha c = -1$. Thus the centre has value -1 and the leaf values are $\{1, i, -i\}$. Independent leaves adjacent to a common vertex must lie in three distinct coordinate directions. Such a claw exists exactly when $d \geq 3$, and all such configurations are conjugate under $\operatorname{Aut} H(d, 3)$.

Lemma 3.3 (The sixth-trace defect)**.** *For the formal claw candidate just described,*

$$\operatorname{tr}(A_d + D)^6 - \operatorname{tr} A_d^6 = -12(2d - 5). \tag{10}$$

Proof. Group the words of length six by cyclic rotation. After the all-A word is removed, the exact contributions are as follows. They can be checked from the distance-walk recurrence

$$w_{\ell+1}(h) = h\, w_\ell(h-1) + h\, w_\ell(h) + 2(d-h) w_\ell(h+1), \qquad w_0(h) = \mathbf{1}_{h=0},$$

where $w_\ell(h) = (A_d^\ell)_{xy}$ whenever the Hamming distance from x to y is h, together with multiplication on the four supported vertices. The entries follow

TABLE 2. Cyclic word contributions for the formal claw.

word	mult.	trace	word	mult.	trace
A^5D	6	0	A^4D^2	6	0
A^3DAD	6	$-12d+6$	A^2DA^2D	3	6
A^3D^3	6	0	A^2DAD^2	6	2
A^2D^2AD	6	2	$ADADAD$	2	0
A^2D^4	6	$8d$	$ADAD^3$	6	-2
AD^2AD^2	3	-2	AD^5	6	0
D^6	1	0			

by multiplying on the four supported vertices, using (3)–(4) and the fact that the three leaves occupy distinct coordinate directions. Multiplicity times trace, summed over Table 2, is $-12(2d-5)$. The ancillary verifier performs this calculation symbolically in d from the displayed distance recurrence and asserts all thirteen rows separately. □

Proposition 3.4 (Sharp five-moment classification)**.** *Let $d \geq 3$ and $0 < |\operatorname{supp} D| \leq 4$. Then*

$$\operatorname{tr}(A_d + D)^k = \operatorname{tr} A_d^k \qquad (1 \leq k \leq 5)$$

if and only if the support is an induced claw and, after permuting its leaves, the values are

$$D_{\text{centre}} = -1, \qquad (D_{\text{leaf},1}, D_{\text{leaf},2}, D_{\text{leaf},3}) = (1, i, -i).$$

Every such shift has sixth-trace defect $-12(2d-5) \neq 0$. *In particular, order six is sharp for detecting all nonzero shifts of support at most four by consecutive power traces.*

Proof. Necessity is Lemma 3.1, Table 1, and the calculation following that table. Conversely, the displayed claw has $p_1 = p_2 = p_3 = 0$, $E = -c^2$, $T = -2c^3$, and $\alpha c = -1$, so Lemma 3.2 verifies the first five traces. The last assertion is Lemma 3.3. □

Proof of Theorem 1.1. Lemma 3.1 excludes support at most three in every dimension. For $d = 1$ this exhausts the three-vertex cell. For $d = 2$, the neighbours of a vertex induce two disjoint copies of K_2, one for each coordinate direction, so they contain no independent three-set and hence no induced claw. Table 1 therefore excludes support four. For $d \geq 3$, the same table leaves only the formal claw and Lemma 3.3 excludes it because $-12(2d-5) \neq 0$. □

4. Distance resolvents of period-three Hamming graphs

For $0 \leq j, h \leq d$, define the ternary Krawtchouk polynomial

$$K_j^{(d)}(h) = \sum_{\ell=0}^{j} (-1)^\ell 2^{j-\ell} \binom{h}{\ell} \binom{d-h}{j-\ell}, \tag{11}$$

with the usual convention for binomial coefficients outside their range, and put $\theta_j = 2d - 3j$.

Proposition 4.1 (Hamming resolvent)**.** *For vertices* $a, b \in \mathcal{F}_d$ *at Hamming distance* h*,*

$$((xI - A_d)^{-1})_{ab} = 3^{-d} \sum_{j=0}^{d} \frac{K_j^{(d)}(h)}{x - \theta_j}. \tag{12}$$

In particular, after multiplication by the minimal polynomial

$$m_d(x) = \prod_{j=0}^{d} (x - \theta_j),$$

the matrix

$$N_d(x) = m_d(x)(xI - A_d)^{-1}$$

has polynomial entries over $\mathbf{Q}$*, each determined only by* h*.*

Proof. The characters of $\mathcal{F}_d$ diagonalize A_d. A character with exactly j nontrivial coordinates has eigenvalue θ_j; summing the rank-one character projectors of that weight gives kernel $3^{-d} K_j^{(d)}(h)$. The spectral decomposition yields (12). The eigenvalues $\theta_0, \ldots, \theta_d$ are distinct, so their product is the minimal polynomial and clears every denominator. □

4.1. **The case $H(3,3)$.** From now on $d = 3$ and $A = A_3$. Its distinct eigenvalues are $6, 3, 0, -3$, so its minimal polynomial is

$$m(x) = x(x-6)(x-3)(x+3). \tag{13}$$

For vertices $a, b \in \mathbf{F}_3^3$, let $\delta(a,b)$ be their Hamming distance and define

$$\begin{aligned} n_0(x) &= x^3 - 6x^2 - 3x + 24, \\ n_1(x) &= x(x-5), \\ n_2(x) &= 2(x-3), \\ n_3(x) &= 6. \end{aligned} \tag{14}$$

Let $N(x)$ be the 27-by-27 matrix with $N_{ab}(x) = n_{\delta(a,b)}(x)$.

Proposition 4.2 (Distance resolvent)**.** *One has the polynomial matrix identity*

$$(xI - A)N(x) = m(x)I. \tag{15}$$

Consequently

$$(xI - A)^{-1}_{ab} = \frac{n_{\delta(a,b)}(x)}{m(x)}$$

away from the spectrum of A.

Proof. Both sides lie in the Bose–Mesner algebra of the Hamming association scheme. Multiplication by A counts one-step changes from each distance class. Substitution of the four polynomials in (14) into these four distance recurrences gives (15). The ancillary verifier also checks all 27^2 entries symbolically over $\mathbf{Z}[x]$ before any support calculation begins. □

5. A finite Rabinowitsch ideal for a prescribed support

Fix $S \subset \mathbf{F}_3^3$ of size r. Let U_S be the 27-by-r coordinate embedding, write $D_S = \operatorname{diag}(v_0, \dots, v_{r-1})$, set $D = U_S D_S U_S^{\mathsf{T}}$, and let $N_S(x) = U_S^{\mathsf{T}} N(x) U_S$.

Proposition 5.1 (Determinant-lemma reduction)**.** *The zero-fibre matrices $A + D$ and A have the same characteristic polynomial if and only if the following polynomial vanishes identically in x:*

$$F_S^{(r)}(x; v) = \sum_{\varnothing \neq T \subseteq \{0,\dots,r-1\}} (-1)^{|T|} m(x)^{r-|T|} \left(\prod_{j \in T} v_j \right) \det N_S(x)[T,T]. \tag{16}$$

Proof. For generic x, the matrix determinant lemma and Proposition 4.2 give

$$\frac{\det(xI - A - D)}{\det(xI - A)} = \det\left(I_r - D_S \frac{N_S(x)}{m(x)} \right).$$

Expand the determinant by principal minors, subtract 1, and multiply by $m(x)^r$. This is (16). Since both sides are rational functions, the resulting identity is also sufficient. □

Let $I_S^{(r)}$ be the ideal over $\mathbf{Q}$ generated by the coefficients in x of $F_S^{(r)}$ after the trace equation

$$v_{r-1} = -(v_0 + \cdots + v_{r-2}) \tag{17}$$

has been substituted. Exact support is imposed by a new variable u and the Rabinowitsch equation

$$u \prod_{j=0}^{r-1} v_j - 1 = 0, \tag{18}$$

again after (17). Denote the resulting extension ideal by

$$J_S^{(r)} = I_S^{(r)} + \left(u \prod_{j=0}^{r-1} v_j - 1 \right) \subset \mathbf{Q}[u, v_0, \ldots, v_{r-2}]. \tag{19}$$

Corollary 5.2 (Exact unit-ideal criterion)**.** *There is no complex zero-fibre isospectral diagonal shift with exact support S if and only if $J_S^{(r)} = (1)$. Equivalently, any reduced Gröbner basis of $J_S^{(r)}$ is $\{1\}$.*

Proof. Equation (18) is solvable precisely when every v_j is nonzero. Proposition 5.1 identifies the remaining coefficient equations with isospectrality. The assertion is therefore the weak Nullstellensatz and the standard unit-ideal criterion for a Gröbner basis [2]. □

The matrix N_S depends, up to simultaneous row and column permutation, only on the abstract pairwise Hamming-distance matrix of S. Thus one unit-ideal calculation per distance type is exhaustive. This reduction is valid specifically at zero quasimomentum; it must not be transferred to a nonzero Floquet phase without a new argument.

Proposition 5.3 (Finite exact decidability)**.** *For every fixed pair (d, r) with $1 \le r \le 3^d$, the existence of an isospectral diagonal shift of $A(H(d, 3))$ with exact support r is decidable by a finite exact computation over $\mathbf{Q}$, with one Rabinowitsch-extended coefficient ideal for each realizable r-point Hamming-distance type.*

Proof. There are finitely many r-subsets of $\mathcal{F}_d$ and hence finitely many distance types. Proposition 4.1 supplies a common rational resolvent denominator and a numerator kernel determined by the distance matrix. The determinant-lemma construction above therefore gives one ideal $J_S^{(r)}$ over $\mathbf{Q}$ per type. Exact-support existence for that type is equivalent to this ideal being proper by Corollary 5.2; a terminating Gröbner-basis computation decides the question. This is a decidability statement, not a claim of practical complexity uniformly in (d, r). □

6. Exact exclusion of support five

Lemma 6.1 (Five-point completeness)**.** *The* $\binom{27}{5} = 80{,}730$ *five-subsets form* 105 *orbits under*

$$\operatorname{Aut} H(3,3) \cong (S_3)^3 \rtimes S_3, \qquad |\operatorname{Aut} H(3,3)| = 1296,$$

and these orbits determine exactly 104 *abstract pairwise Hamming-distance types.*

Proof. The primary verifier enumerates all five-subsets and minimizes the ten-entry upper-triangular distance word over all 5! point relabelings. Independently, the completeness auditor partitions all subsets by union-find under eight generators of $(S_3)^3 \rtimes S_3$, obtains 105 orbits, and only then canonicalizes their distance matrices. Both routes give 104 types and the same serialized-key digest displayed below. The passage from 105 orbits to 104 distance types is valid because $N_S(x)$ is determined entirely by the pairwise Hamming-distance matrix. □

For every type the verifier constructs (16), makes the exact substitution $v_4 = -(v_0 + \cdots + v_3)$, appends (18), and computes a Gröbner basis over $\mathbf{Q}$ in the variables $u, v_0, \ldots, v_3$ using graded reverse lexicographic order.

Proposition 6.2 (Support-five exclusion)**.** *All* 104 *Rabinowitsch extension ideals have Gröbner basis* $\{1\}$*. Hence there is no exact-support-five zero-fibre isospectral shift on* $H(3,3)$*.*

Exact computer-assisted proof. Lemma 6.1 makes the type list exhaustive. For each type, the verifier constructs $F_S^{(5)}$, extracts every coefficient in x, performs the trace substitution, and constructs $J_S^{(5)}$ from (19). Exact arithmetic over $\mathbf{Q}$ returns the reduced basis $\{1\}$ for each of the 104 ideals. Corollary 5.2 excludes exact support five for every type. □

The sorted distance keys, serialized as comma-separated entries with one key per line, have SHA-256 digest

`222cdf7f5c212970fb39bf5e723bfcf29dfdfa72c6bd77ab9f25c10e5686cea9`.

The reference run used Python 3.12.13 and SymPy 1.14.0 and returned 104/104 unit ideals.

7. Exact exclusion of support six

Lemma 7.1 (Six-point completeness)**.** *The six-point support counts are those in Table 3. Exactly two nonconjugate automorphism orbits share one abstract distance matrix; all other distance types contain one orbit.*

Proof. One exact computation partitions all 296,010 subset masks by union-find under eight generators of the automorphism group. A separate Burnside sum over all 1,296 group elements gives

$$\frac{1}{1296} \sum_{g \in \operatorname{Aut} H(3,3)} \#\operatorname{Fix}_{\binom{\mathcal{F}_3^3}{6}}(g) = \frac{416{,}016}{1{,}296} = 321,$$

TABLE 3. Exact combinatorial counts for six-point supports.

quantity	*exact value*
six-subsets	$\binom{27}{6} = 296{,}010$
automorphism group order	1,296
automorphism orbits	321
pairwise-distance types	320

independently confirming completeness. Canonicalizing the fifteen-entry distance words of the 321 orbit representatives gives 320 types, with one distance word occurring twice. The two representatives and their distinct coordinate-fibre profiles are recorded by the ancillary auditor. □

The distinction in the final two rows is essential. Exactly two nonconjugate automorphism orbits share one abstract distance matrix. They may be coalesced here only because the resolvent kernel in Proposition 4.2 is distance-dependent.

For each of the 320 types, the support-six verifier repeats the construction of Section 6 with $v_5 = -(v_0 + \cdots + v_4)$.

Proposition 7.2 (Support-six exclusion)**.** *All* 320 *Rabinowitsch extension ideals have Gröbner basis* $\{1\}$ *over* $\mathbf{Q}$*. Hence no zero-fibre isospectral diagonal shift on* $H(3,3)$ *has exact support six.*

Exact computer-assisted proof. Lemma 7.1 makes the distance-type list exhaustive. For every type, the support-six verifier constructs $F_S^{(6)}$, extracts all coefficients in x, performs the trace substitution, and constructs $J_S^{(6)}$. All 320 exact reduced bases over $\mathbf{Q}$ are $\{1\}$. Corollary 5.2 therefore excludes exact support six. □

The exact JSONL computation transcript records one header, 320 sorted type records and one summary record. Its principal digests are The ref-

TABLE 4. Digests for the support-six release. The first two are deterministic mathematical digests; the last is an archival file-integrity digest.

object	SHA-256
sorted distance keys	`b7a0ff2f436adb2f9eed342cf33324b4 45cb118672017d5e8eb6beb619037ecb`
canonical transcript	`192dddaf12e4622821aec0f58ff4b95e d46e853f14333c44a665c3fb0d0ada1e`
complete JSONL file	`c0c64dc9f4ca7bfd8b63cbbcd08ea47 368948aeb31917067cf1397d9f86c7257`

erence run returned 320/320 unit ideals and no nonunit index. Types 1, 90, 160 and 320 were recomputed in fresh processes and reproduced their

stored ideal hashes. A separate union-find/Burnside program reproduced the 321 orbits, the 320 types and the distance-key digest. The complete-file digest is expected to change on a fresh run because the log includes process identifiers, platform data and timings; it authenticates the distributed file only. The canonical transcript digest omits those ephemeral fields and is the reproducible comparison target.

Proof of Theorem 1.2. Theorem 1.1 excludes support at most four. Propositions 6.2 and 7.2, through Corollary 5.2, exclude exact supports five and six. □

Proof of Corollary 1.3. Full Floquet isospectrality implies equality of the characteristic polynomials at zero quasimomentum, so Theorem 1.2 applies. □

8. Product transfer and cylinder potentials

Let $1 \le k < d$ and let V_0 be a $(3\mathbf{Z})^k$-periodic potential. Its *cylinder lift* to $\mathbf{Z}^d$ is

$$V(n_1, \dots, n_d) = V_0(n_1, \dots, n_k).$$

Proposition 8.1 (Cylinder equivalence)**.** *The cylinder lift V is Floquet isospectral to zero on $\mathbf{Z}^d$ if and only if V_0 is Floquet isospectral to zero on $\mathbf{Z}^k$.*

Proof. Write $z = (z', z'')$. In the product basis the Floquet matrix is the Kronecker sum

$$L_V^{(d)}(z) = L_{V_0}^{(k)}(z') \otimes I + I \otimes L_0^{(d-k)}(z'').$$

If the two k-dimensional matrices are isospectral for every z', their Kronecker sums with the same second factor are isospectral, proving one direction. Conversely, equality of the two d-dimensional characteristic polynomials implies, for every integer $m \ge 1$, equality of power traces and hence of trace exponentials near $t = 0$. Using

$$\operatorname{tr} e^{t(X \otimes I + I \otimes B)} = (\operatorname{tr} e^{tX})(\operatorname{tr} e^{tB}),$$

the common factor $\operatorname{tr} e^{tB}$ can be cancelled near zero because its value at $t = 0$ is 3^{d-k}. Thus the trace exponentials, and therefore the characteristic polynomials, of $L_{V_0}^{(k)}(z')$ and $L_0^{(k)}(z')$ agree for every z'. □

Corollary 8.2 (Sparse cylinder obstruction)**.** *If $d \ge 3$ and a nonzero $(3\mathbf{Z})^d$-periodic Floquet-isospectral potential is, up to a coordinate permutation, a cylinder lift from three coordinate factors, then its support in a fundamental cell satisfies*

$$|\operatorname{supp} V| \ge 7 \cdot 3^{d-3}.$$

A cylinder depending on at most two coordinates must be zero.

Proof. For $d = 3$ the first assertion is Corollary 1.3. For $d > 3$, Proposition 8.1 and that corollary give $|\operatorname{supp} V_0| \geq 7$, and the lift repeats each supported value 3^{d-3} times. The assertion for at most two active coordinates follows from the small-period rigidity part of [1, Theorem C]. □

9. Reproducibility protocol

The ancillary files are:

- `verify_floquet_333_sparse.py`, for the support-four trace table, the symbolic sixth-trace calculation, the resolvent identity and all support-five ideals;
- `verify_floquet_333_support6.py`, for the orbit reduction, all support-six ideals, the JSONL transcript and deterministic comparison with a stored reference run;
- `audit_h333_support6_completeness.py`, for the independent union-find and Burnside audit;
- `floquet_support6_certificate.jsonl`, the complete hashed support-six computation transcript;
- `requirements.txt`, pinning the symbolic-algebra dependency;
- the ancillary manifest, for release-file hashes and the reference environment.

The three principal commands are

```
python verify_floquet_333_sparse.py --quiet
python audit_h333_support6_completeness.py \
  --show-collision --cross-check-five
python verify_floquet_333_support6.py --workers 6 \
  --output /tmp/floquet_support6_check.jsonl --force \
  --reference floquet_support6_certificate.jsonl
```

The reference environment was Python 3.12.13 with SymPy 1.14.0 [9]. One reference run took about 240 seconds for support five and 912 seconds for support six; runtimes are not part of the result. Mathematical verification consists of checking the symbolic trace and resolvent identities, reproducing the orbit and type counts, recomputing every reduced basis, and comparing every deterministic type record with the stored transcript. The comparison deliberately ignores process identifiers, platform strings and timings. The JSONL file is an exact computation transcript, not a standalone formal Nullstellensatz certificate containing multiplier identities; independent verification therefore reruns the Gröbner computations.

10. Scope and open directions

The exact status is:

Table 5. Proved and open strata.

stratum	status
support at most 3, every $d \geq 1$	impossible by traces 1–3
support 4, every $d \geq 1$	impossible by traces through order 6
support 5 in $(3\mathbf{Z})^3$	exact computer-assisted impossibility
support 6 in $(3\mathbf{Z})^3$	exact computer-assisted impossibility
nonzero periodic-fibre shift on $H(3,3)$	exists, necessarily support ≥ 7
exact support 7 in $H(3,3)$	open
unrestricted $(3,3,3)$ existence	open

The nearest finite continuation is support seven, but its orbit count and seven-variable Rabinowitsch ideals are materially larger. A potentially more global approach is to combine several roots-of-unity quasimomenta. The present proof deliberately uses only $z = 1$; a few nonzero phases may impose incompatible low-rank resolvent identities simultaneously and could obstruct many support sizes at once. Neither continuation is asserted here. Nor does the $d = 3$ computation exclude arbitrary supports five or six in dimensions $d > 3$; only the cylinder consequence in Corollary 8.2 is claimed across dimensions.

Data and code availability

The exact scripts and the full JSONL transcript are supplied with the source archive accompanying this manuscript. No proprietary solver or floating-point computation is required.

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